Linear Algebra - 24 DS

Problem Sheet: Subspaces, Bases and Dimension

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A Subspaces

1. **($\mathbb{R}^3$ - Line through origin)** Determine whether

$$W_1 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; x = 2t,\ y = -t,\ z = 3t,\ t \in \mathbb{R} \right\}$$

is a subspace of $\mathbb{R}^3$. If yes, find its dimension and geometric description.

2. **($\mathbb{R}^3$ - Plane through origin)** Check if

$$W_2 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; 2x - y + 3z = 0 \right\}$$

is a subspace of $\mathbb{R}^3$. Find a basis and state the dimension.

3. **($\mathbb{R}^3$ - Not containing zero)** Is

$$W_3 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; x + y + z = 1 \right\}$$

a subspace of $\mathbb{R}^3$? Justify your answer.

4. **($\mathbb{R}^4$ - Hyperplane)** Show that

$$W_4 = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 \;\middle|\; x_1 + x_2 - x_3 + 2x_4 = 0 \right\}$$

is a subspace of $\mathbb{R}^4$. Find its dimension and a basis.

5. **($\mathbb{R}^4$ - Span of vectors)** Let

$$W_5 = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

   Is W_5 a subspace of $\mathbb{R}^4$? Justify and find its dimension.

6. **(Conceptual)** Explain why every subspace of $\mathbb{R}^n$ must contain the zero vector. Give an example of a subset of $\mathbb{R}^3$ that contains the zero vector but is NOT a subspace.

B Basis of Various Subspaces

1. **(Basis from span in $\mathbb{R}^3$)** Find a basis for

$$W = \operatorname{span}\left\{ \begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}2\\4\\6\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix} \right\}$$

2. **(Basis for a plane in $\mathbb{R}^3$)** Find a basis for the subspace

$$W = \left\{ \begin{bmatrix}x\\y\\z\end{bmatrix} \in \mathbb{R}^3 \;\middle|\; x - y + 2z = 0 \right\}$$

3. **(Basis for a line in $\mathbb{R}^4$)** Find a basis for

$$W = \left\{ \begin{bmatrix}2t\\-t\\3t\\t\end{bmatrix} \;\middle|\; t \in \mathbb{R} \right\} \subseteq \mathbb{R}^4$$

4. **(Basis from column space)** Find a basis for the subspace spanned by the columns of

$$A = \begin{bmatrix}1 & 2 & 3 & 4\\2 & 4 & 6 & 8\\1 & 0 & 1 & 2\end{bmatrix}$$

5. **(Extending to a basis)** Extend the linearly independent set

$$\left\{ \begin{bmatrix}1\\1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\\0\end{bmatrix} \right\}$$

   to a basis of $\mathbb{R}^4$.

6. **(Conceptual)** Can a set of 4 vectors in $\mathbb{R}^3$ form a basis for a subspace? If yes, what is the maximum possible dimension? Explain with an example.

C Dimensions of Subspaces

1. **(Dimension of a plane in $\mathbb{R}^3$)** Find the dimension of

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; 3x - y + 2z = 0 \right\}$$

   Also, describe the geometric shape.

2. **(Dimension of span in $\mathbb{R}^4$)** Find the dimension of

$$W = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

3. **(Dimension of solution space in $\mathbb{R}^4$)** Find the dimension of the solution space of

$$x_1 + x_2 - x_3 = 0, \quad 2x_1 - x_2 + x_4 = 0$$

   (in $\mathbb{R}^4$).

4. **(Dimension of intersection)** Find the dimension of $W_1 \cap W_2$ where

$$W_1 = \operatorname{span}\{(1,0,0),(0,1,0)\}, \quad W_2 = \operatorname{span}\{(1,1,0),(0,0,1)\}$$

   in $\mathbb{R}^3$.

5. **(Dimension comparison)** Let W_1 be a plane through origin in $\mathbb{R}^3$ and W_2 be a line through origin in $\mathbb{R}^3$ not contained in W_1. What is $\dim(W_1 \cap W_2)$? Justify.

6. **(Conceptual)** Is it possible for a subspace of $\mathbb{R}^4$ to have dimension 5? Why or why not? Explain the relationship between dimension and the ambient space.